

Data Output

Data Output

While its nice to be able to read in a variety of data formats, it's equally important to be able to output data somewhere.

The `readr` package in the tidyverse provides data exporting functions which have the pattern `write_*`:

- `write_csv()`,
- `write_delim()`, others.

From `write_csv()` documentation:

```
write_csv(x, file,  
  na = "NA", append = FALSE,  
  col_names = !append, quote_escape = "double",  
  eol = "\n", path = deprecated()  
)
```

Data Output

`x`: data frame you want to write

`file`: file path where you want to R object written; it can be:

- an absolute path,
- a relative path (relative to your working directory),
- a file name only (which writes the file to your working directory)
- remember to include the file extension (`.csv`, `.txt`, or `.tsv`)

Examples

```
write_csv(dat, file = "YouthTobacco_newNames.csv")
```

```
write_delim(dat, file = "YouthTobacco_newNames.csv", delim = ",")
```

GUT CHECK!

What does `write_csv()` do? Saves data to:

- A. R's memory
- B. A file on your hard drive
- C. A ggplot

R binary file

.rds is an extension for R native file format.

`write_rds()` and `read_rds()` from `readr` package can be used to write/read a single R object to/from file.

Saving datasets in .rds format can save time if you have to read it back in later.

```
# write an object: a data frame "dat"  
write_rds(dat, file = "yts_dataset.rds")
```

```
# write an object: vector "x"  
x <- c(1, 3, 3)  
write_rds(x, file = "my_vector.rds")
```

```
# read an object from file and assign to a new object named "y"  
x2 <- read_rds(file = "my_vector.rds")  
x2
```

```
[1] 1 3 3
```

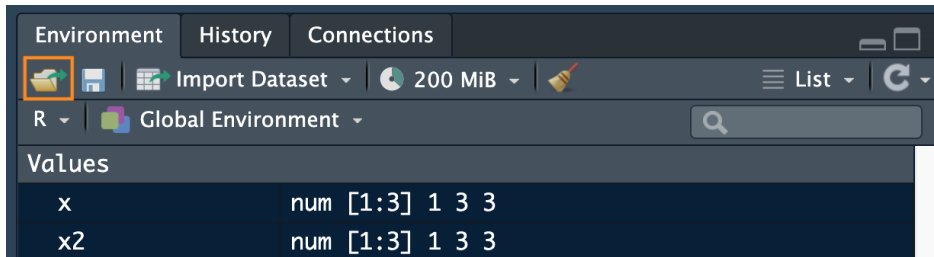
Saving multiple objects

You may want to export a set of objects from R for later use, either to save time or to use in another R script. You can output these to an `.RData` file individually, or save your entire Environment with `save.image()`.

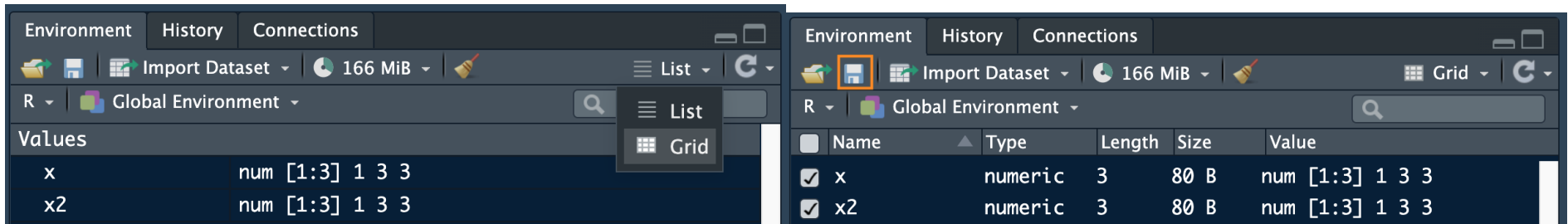
```
save(x, x2, file = "x_x2_output.RData")  
save.image(file = "my_environment.RData")
```

Using RStudio for importing/exporting data

If there is an `.rds` or `.RData` file that you want to work with, you can open it into your environment using the file icon.



Can also save your entire environment or a subset of objects in your environment to a new `.RData` file with the save icon. Click the “List” button and switch to “Grid” to select which objects to delete or keep before saving the Environment.



REMINDER: Saving a ggplot to file

A few options:

- RStudio > Plots > Export > Save as image / Save as PDF
- RStudio > Plots > Zoom > [right mouse click on the plot] > Save image as
- In the code

```
ggsave(filename = "saved_plot.png", # will save in working directory
        plot = rp_fac_plot,
        width = 6, height = 3.5)      # by default in inches
```


Summary

- Use `write_csv()` and `write_delim()` from the `readr` package to write your (modified) data
- `.rds` files can be handy for saving intermediate work
- Can save environment (or subset) using `save()` and `save.image()`

Resources & Lab

- ▮ [Class Website](#)
- ▮ [Data Output Lab](#)
- ▮ [Posit's Data Import Cheatsheet](#)
- ▮ [Day 2 Cheatsheet](#)



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